Solution of Quiz # 1

Q1 . Suppose you have a transportation network with 10 cities represented by the adjacency matrix below. Each connection between cities has a specific weight (representing distance in kilometers). The goal is to analyze paths between City A and City J using different search algorithms.

	A	B	C	D	E	F	G	H	I	J
A	0	2	4	0	0	0	0	0	0	0
B	2	0	0	7	0	3	0	0	0	0
C	4	0	0	1	5	0	0	0	0	0
D	0	7	1	0	0	0	2	6	0	0
E	0	0	5	0	0	0	0	0	4	0
F	0	3	0	0	0	0	8	0	0	5
G	0	0	0	2	0	8	0	0	0	3
H	0	0	0	6	0	0	0	0	1	0
I	0	0	0	0	4	0	0	1	0	2
J	0	0	0	0	0	5	3	0	2	0

1. Using BFS, what is the shortest path from City A to City J (considering unweighted paths)?
2. Using DFS, find the longest path from City A to City J without revisiting any city.
3. Using the Greedy Algorithm (always choose the nearest next city), what path would you find from City A to City J? Is this the optimal path? Justify your answer.

Breadth-First Search explores level by level and guarantees the shortest path in terms of the number of edges, not the weights. Adjacency Matrix (Unweighted Interpretation): If there's a non-zero weight, we treat it as a connection (1). Otherwise, it's 0.

Step-by-Step BFS Traversal:

1. Start at A → Queue: `[A]`, Path: `[]`
2. Visit A: Add neighbors B and C → Queue: `[B, C]`, Path: `A`
3. Visit B: Add neighbors D and F → Queue: `[C, D, F]`
4. Visit C: Add neighbors D and E → Queue: `[D, F, D, E]`
5. Visit D: Add neighbors G and H → Queue: `[F, D, E, G, H]`
6. Visit F: Add neighbors G and J → Queue: `[D, E, G, H, G, J]`
7. J is found.

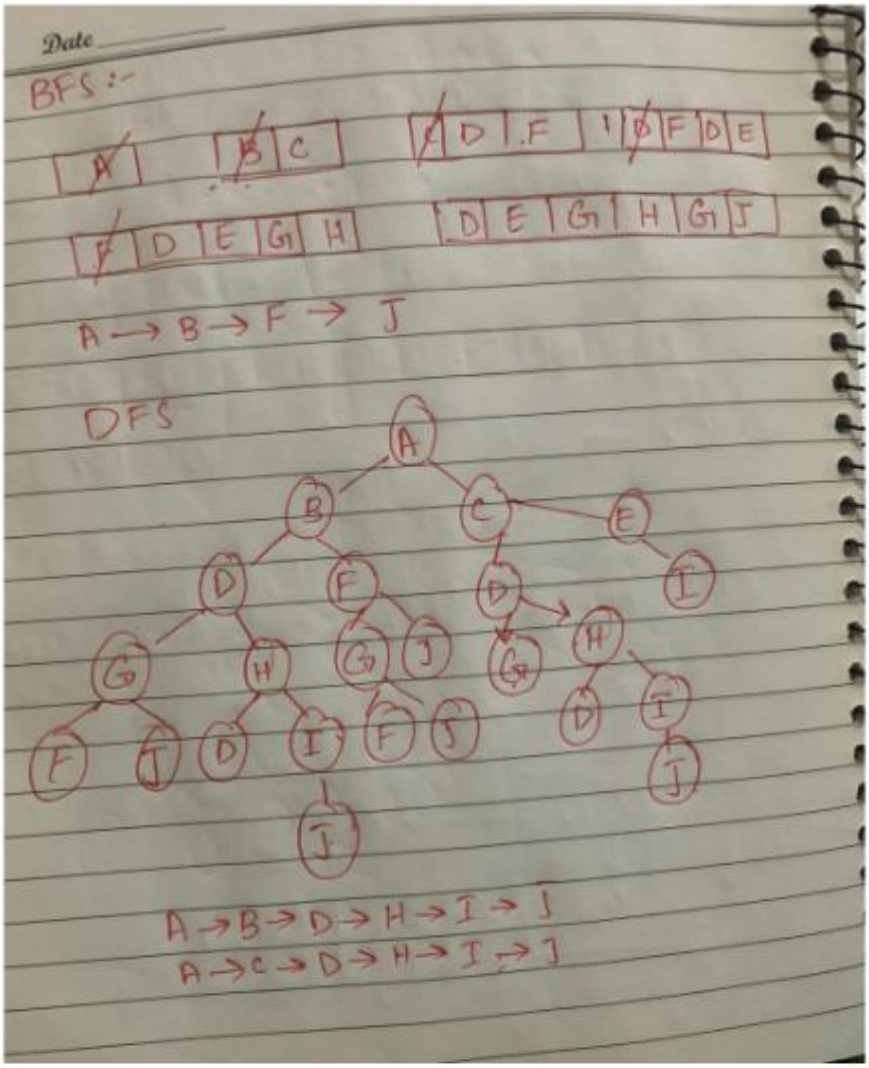
Shortest Path: A → B → F → J (3 edges)

Step-by-Step DFS Traversal:

1. Start at A → Path: `A`
2. Go to B → Path: `A → B`
3. Go to D → Path: `A → B → D`
4. Go to H → Path: `A → B → D → H`
5. Go to I → Path: `A → B → D → H → I`
6. Go to J → Path: `A → B → D → H → I →`

Path Length: 5 edges

Longest Path Found: A → B → D → H → I → J



Step-by-Step Greedy Approach:

1. Start at A:
   - B (2 km) or C (4 km) → Choose B (shorter).
2. At B:
   - D (7 km) or F (3 km) → Choose F (shorter).
3. At F:
   - G (8 km) or J (5 km) → Choose J (shorter).

Greedy Path: A → B → F → J

Total Distance: 2 + 3 + 5 = 10 km

Is this the Optimal Path?

Let's check for a better path manually:

- A → C → D → G → J
  - 4 + 1 + 2 + 3 = 10 km (Same total distance)

the Greedy Algorithm found an optimal path. However, Greedy algorithms are not guaranteed to always find the optimal path in all scenarios.

Q.2 Solve the following by using the Hill climbing algorithm.

2	3	
1	8	4
7	6	5

Initial state

1	2	3
8		4
7	6	5

Goal state

Date

R

1	2	3
	4	6
7	5	8

S

1	2	3
4	5	6
7	8	

G

no of missed place tiles. 3

1, 2, 3 on Place
4 (1), 5 (2), 8 (

1	2	3
4	6	
7	5	8

h = 4

1	2	3
7	4	6
	5	8

h = 4

1	2	3
4		6
7	5	8

R

1	2	3
	4	6
7	5	8

3

1	2	3
4	5	6
7		8

(1)

1		3
4	2	6
7	5	8

(3)

1	2	3
4	5	6
7	8	

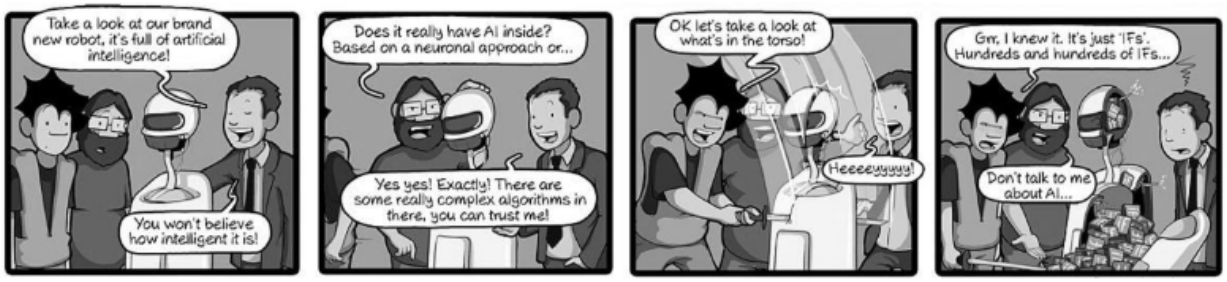
(0)

1	2	3
4	5	6
	7	8

(1)

No.

Q1. Consider the comic below and answer the questions that follow: -



A. Which dimension of AI definitions best matches the context in the above comic? Give strong reasons.

MODEL ANSWER:

Thinking Rationally: 1. No mention of action consequences or achieving goal or behaviour. 2. Only mention of complex algorithms but no mention of thinking like a human.

RUBRICS:

1 mark : getting the dimension right

2 marks: for valid reasons.

NOTE: There can be other reasons to justify the answer. If they are correct and support the choice the student has made he can be awarded marks accordingly.

B. What type of Agent did the robot turn out to be? Why were the guys disappointed to find many IFs? Isn't it AI?.

MODEL ANSWER:

It is a simple reflex agent. Although it falls under AI it has limited intelligence and ignores all previous percepts and needs a fully observable.

RUBRICS:

1.5 marks for mentioning simple reflex agent

1.5 marks for the rest of the answer.

C. Give PEAS for solving a jigsaw puzzle.

MODEL ANSWER:

Performance Measure: Correctly assembling the puzzle pieces to form the complete image. Minimizing the time taken to solve the puzzle. Ensuring accuracy (no misplaced pieces). Efficiently exploring possible placements of pieces.

Environment: The puzzle board where pieces need to be placed. The set of jigsaw pieces with unique shapes and colors. A digital or physical workspace where the AI operates. Constraints such as the total number of pieces and edge pieces.

Actuators: If digital: Algorithms that move, rotate, and place pieces in a virtual environment. If physical: Robotic arms or mechanisms that pick and place puzzle pieces.

Sensors: If digital: Image recognition and pattern-matching algorithms to analyze pieces. If physical: Cameras, vision systems, or touch sensors to identify piece shapes, colors, and edges.

RUBRICS:

1 mark for getting the PEAS full form correct.

0.5 marks for each of P E A S.

NOTE: Students only need to either give PEAS for either Physical Version or Digital Version.

D. Which of the categories of Nature of environments does solving jigsaw puzzles belong to? Cover all the characteristics of the task environment. (**HINT**: fully vs partially observable, deterministic vs nondeterministic etc.)

MODEL ANSWER:

Fully Observable, Deterministic (if digital version else can also be Non-deterministic), Sequential, Static, Discrete, Single Agent (Benign)

RUBRICS:

0.5 marks for each

Q1: For each of the following activities, give a PEAS description of the task environment. (Any 2)

1. Playing soccer

- Performance Measure: Goals scored, passes completed, successful tackles, overall team performance, adherence to rules.
- Environment: Soccer field, teammates, opponents, referee, spectators, weather conditions.
- Actuators: Legs for running, kicking, and tackling; arms for balance; communication with teammates.
- Sensors: Eyes for vision (ball, teammates, opponents), ears for hearing coach and referee, proprioception for body movement awareness.

2. Exploring the subsurface of the Arabian Sea

- Performance Measure: Area explored, data accuracy, obstacle avoidance, efficiency of movement, energy consumption.
- Environment: Underwater terrain, marine life, ocean currents, water pressure, temperature.
- Actuators: Propellers or thrusters for movement, robotic arms for sampling, ballast system for depth control.
- Sensors: Sonar, cameras, pressure sensors, temperature sensors, chemical detectors.

3. Shopping for used AI books on the Internet

- Performance Measure: Price comparison, book availability, delivery options, accuracy of search results.
- Environment: Online bookstores, e-commerce platforms, user interface, internet connection.
- Actuators: Mouse clicks, keyboard input, voice commands (if applicable). • Sensors: Screen display, website feedback, recommendation algorithms, user input tracking.

4. Playing a tennis match

- Performance Measure: Points won, accuracy of shots, unforced errors, fitness level, strategy execution.
- Environment: Tennis court, opponent, net, ball, weather (if outdoors). • Actuators: Arms for racket swings, legs for movement, torso for balance, communication for calls (if doubles).
- Sensors: Eyes for ball tracking, ears for opponent cues, proprioception for movement coordination.

5. Practicing tennis against a wall

- Performance Measure: Number of successful returns, consistency, ball control, stamina.
- Environment: Wall, ball, racket, ground surface, weather (if outdoors). • Actuators: Arms for swinging, legs for positioning, torso for balance. • Sensors: Eyes for ball tracking, proprioception for movement awareness, ears for ball impact sounds.

Q2:.

a) Autonomous Delivery Drone:

Scenario: Imagine an autonomous drone designed to deliver packages in a city. The drone has sensors like GPS, cameras, and altitude sensors, and it can take actions such as moving up/down, navigating left/right, or dropping packages at specific locations.

Question:

- ★ What would the environment look like for this delivery drone?
- ★ What are the drone's sensors and actuators in this scenario?
- ★ How does the drone perceive changes in the environment, and what decisions would it make based on those perceptions?

Answers:

- ★ Environment: Cityscape with buildings, roads, traffic, weather conditions (wind, rain), pedestrians, delivery locations, and obstacles (birds, power lines).

- ★ Sensors: GPS (location tracking), cameras (object detection, navigation), altitude sensors (height control), LiDAR/radar (obstacle detection), wind sensors (weather adaptation).
  Actuators: Propellers (movement), servos (package release), control surfaces (stabilization).

- ★ Perception & Decisions:

  - ○ Uses GPS and cameras to navigate and detect obstacles.
  - ○ Adjusts altitude based on buildings and air traffic.
  - ○ Avoids collisions using LiDAR/radar.
  - ○ Modifies speed/direction based on wind sensors.

b) E-commerce Recommendation System:

Scenario: An e-commerce website uses a recommendation system to suggest products to users based on their past behavior (e.g., purchases, clicks, search history). The recommendation agent adapts to the user's preferences over time to suggest items that are most likely to be purchased.

Question:

- **What would the environment look like for this recommendation agent? • How does the agent perceive changes in the environment (e.g., new user data, product preferences)?**
- **How does the agent decide which products to recommend, and what type of learning process might be involved?**

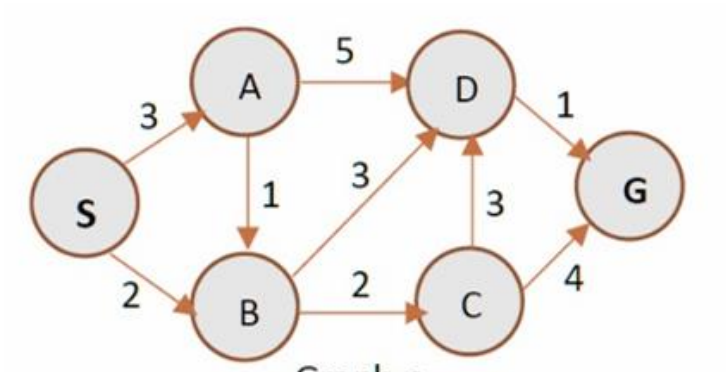
Answers

- **Environment: E-commerce platform, user profiles, product catalog, browsing history, purchase history, search queries, seasonal trends, and competitor pricing.**

- **Perception: Tracks user interactions (clicks, purchases, searches, time spent on pages), detects changing preferences, analyzes trends, and incorporates new products or promotions.**

- **Uses collaborative filtering, content-based filtering, and deep learning models to suggest products. Employs machine learning techniques (supervised, unsupervised, reinforcement learning) to refine recommendations over time.**

Q3: Answer the following questions about the search problem shown above. Break any ties alphabetically. For the questions that ask for a path, please show complete traversal with the help of open and closed lists.



Graph a

(a) What path would uniform cost graph search return for this search problem?
(b) What path would depth-first graph search return for this search problem?
(c) What path would A Graph search, using below given heuristic, returns for this search problem?
(d) Consider the heuristics for this problem shown in the table below.

State	h_1	h_2
S	5	4
A	3	2
B	6	6
C	2	1
D	3	3
G	0	0

i. Is h1 admissible? Yes No
ii. Is h1 consistent? Yes No
iii. Is h2 admissible? Yes No
iv. Is h2 consistent? Yes No

(a) What path would uniform cost graph search return for this search problem?
S → B → D → G (Total cost = 5)
(b) What path would depth-first graph search return for this search problem?
S → A → B → C → G (Not optimal, cost = 10)
(c) What path would A Graph search, using below given heuristic, returns for this search problem?

***Final Path (A* with h1):**
S → B → C → G (Total cost = 8)
***Final Path (A* with h2):**
S → B → C → G (Total cost = 8)

Neither heuristic is admissible or consistent thats why the results using both are not optimal.

Step 1: True optimal costs to goal G

From the graph:

$D \to G = 1 \Rightarrow h^*(D) = 1 \Rightarrow h^*(D) = 1$

$C \to G = 4 \Rightarrow h^*(C) = 4 \Rightarrow h^*(B) = 4$

$B \to C \to G = 2+4 = 6 \Rightarrow h^*(B) = 4$

but better: $B \to D \to G = 3+1 = 4$

$h^*(B) = 4 \Rightarrow h^*(B) = 4$

$A \to D \to G = 5+1 = 6 \Rightarrow h^*(A) = 5$

$A \to B \to D \to G = 1+3+1 = 5 \quad h^*(A) = 5$

$h^*(S) = 6$

State	h_1'
S	6
A	5
B	3
C	2
D	3
G	0

State	h_1'	h_1'
S	5	5
A	3	3
B	6	2
C	2	3
D	3	0
G	0	0

✓ (i) Is h_1' admissible?

Check:

S: $5 \leq 6$ ✓

A: $3 \leq 5$ ✓ $\quad 6 > 4$ ✗ (violates)

B: $2 \leq 4$ ✓ $\quad 2 \leq 4$ ✗

C: $2 \leq 4$ ✓ $\quad 3 > 1$ ✗ (violates)

D: $3 > 1$ ✓ $\quad 0 \leq 0$ ✗

G: $0 \leq 0$ ✓

Since it overestimates at B and D:

✗) Answer NO

✓ (ii) Is h_1' consistent?

Consistency condition:

$h(n) \leq c(n,m) + h(m)$

Key edges:

$B \to C \quad 6 \leq 2+2 = 4$ ✗

$D \to G \quad 3 \leq 1+0 = 1$ ✗

Since violations exist:

✗) Answer (ii): NO

✓ (iii) Is h_2' admissible?

Check:

S: $4 \leq 6$ ✓

A: $2 \leq 5$ ✓ $\quad 6 > 4$ ✗

B: $6 > 4$ ✗ $\quad 3 > 1$ ✗

C: $1 \leq 4$ ✓ $\quad 0 \leq 0$ ✗

✗) Answer (iii): NO

✓ FINAL ANSWERS (Correct)

i) NO	iii) NO
iii) NO	iv) NO